DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.TECH CSE (AIML) - ACADEMIC REGULATIONS & GUIDELINES

1. ATTENDANCE REQUIREMENTS
Every candidate is expected to attend 100% of the classes. To be eligible for semester-end examinations, a stud
Condonation of shortage of attendance between 65% and 75% may be recommended on medical grounds subje
Students with attendance below 65% will be detained and will not be permitted to appear for the semester exami

2. GRADING SYSTEM AND CGPA CALCULATION
Academic performance is evaluated on a 10-point scale:
- Grade O (Outstanding): 90-100% marks, Grade Points: 10
- Grade A+ (Excellent): 80-89% marks, Grade Points: 9
- Grade A (Very Good): 70-79% marks, Grade Points: 8
- Grade B+ (Good): 60-69% marks, Grade Points: 7
- Grade B (Above Average): 55-59% marks, Grade Points: 6
- Grade C (Average): 50-54% marks, Grade Points: 5
- Grade P (Pass): 40-49% marks, Grade Points: 4
- Grade F (Fail): Below 40% marks, Grade Points: 0

Cumulative Grade Point Average (CGPA) is computed as the weighted average of grade points across all compl

3. CORE CSE & AIML CURRICULUM SYLLABUS

CS301: DATA STRUCTURES AND ALGORITHMS (4 CREDITS)
Prerequisites: Problem Solving through C/Python.
Unit I: Linear data structures: Stacks, Queues, Linked Lists (Singly, Doubly, Circular).
Unit II: Non-linear data structures: Binary Trees, Binary Search Trees, AVL Trees, Heaps.
Unit III: Graph algorithms: Breadth-First Search (BFS), Depth-First Search (DFS), Dijkstra's shortest path, Prim's
Unit IV: Sorting and Searching: Quick Sort, Merge Sort, Heap Sort, Binary Search, Hash Tables with collision res

AI402: MACHINE LEARNING & NEURAL NETWORKS (4 CREDITS)
Prerequisites: Linear Algebra, Probability, and Python Programming.
Unit I: Supervised Learning: Linear Regression, Logistic Regression, Decision Trees, Random Forests, Support
Unit II: Unsupervised Learning: K-Means Clustering, Hierarchical Clustering, Principal Component Analysis (PCA
Unit III: Deep Learning Foundations: Feedforward Neural Networks, Backpropagation, Activation functions (ReLU
Unit IV: Optimization: Stochastic Gradient Descent (SGD), Adam Optimizer, Regularization techniques (L1/L2, D
Unit V: Model Evaluation: Precision, Recall, F1-Score, ROC-AUC curve, Cross-Validation strategies.

4. INTERNSHIP AND CAPSTONE PROJECT GUIDELINES

4.1 Mandatory Summer Internships

Students must undergo an industry internship of minimum 6 weeks duration during the summer vacation after th
Internship evaluation carries 2 academic credits evaluated based on industry mentor feedback, technical report,

4.2 Capstone Major Project (8th Semester - 10 Credits)

Students are required to design, implement, and validate an industry-grade problem.
Recommended tracks:

- Artificial Intelligence & Machine Learning
- Cloud-native Applications & Microservices
- Cybersecurity & Distributed Systems
- Data Science & Advanced Analytics

Evaluation involves Phase-1 design defense, continuous progress monitoring, final code audit, and plagiarism ve

5. CAREER PATHWAYS & PLACEMENT ELIGIBILITY

Students maintaining a CGPA of 7.5 and above with zero standing arrears are eligible for tier-1 placement recrui
Target skills for software roles include Data Structures & Algorithms, Object-Oriented Design, and System Archit
For AI and Data Science roles, competency in Python, PyTorch/TensorFlow, SQL, and exploratory data analysis